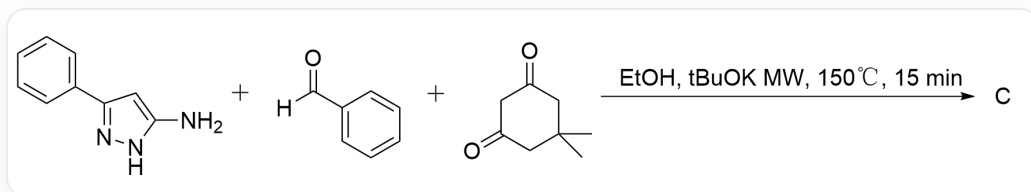


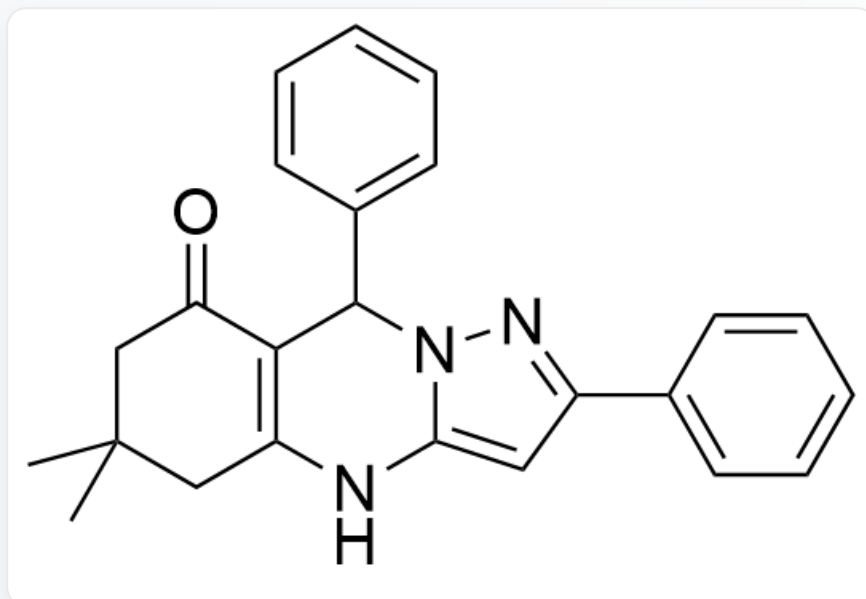
题目



NC1=CC(C2=CC=CC=C2)=NN1.O=C(C1=CC=CC=C1)[H].O=C1CC(C)(C)CC(C1)=O>EtOH, tBuOK
MW, 150°C, 15 min>C

不考虑对映异构体的情况下请预测该反应的产物C（分子式：C₂₄H₂₅N₃O₂），并指出该结构中有几个环

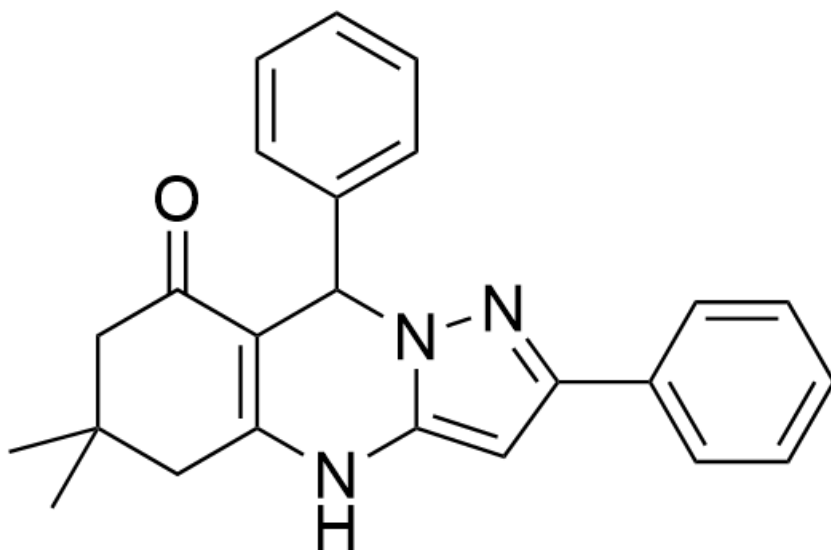
A.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

该结构中有6个环

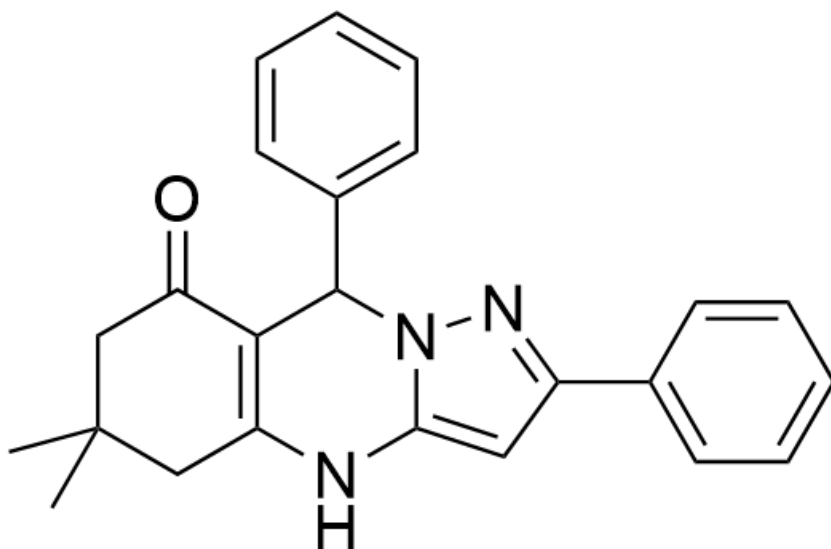
B.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

该结构中有5个环

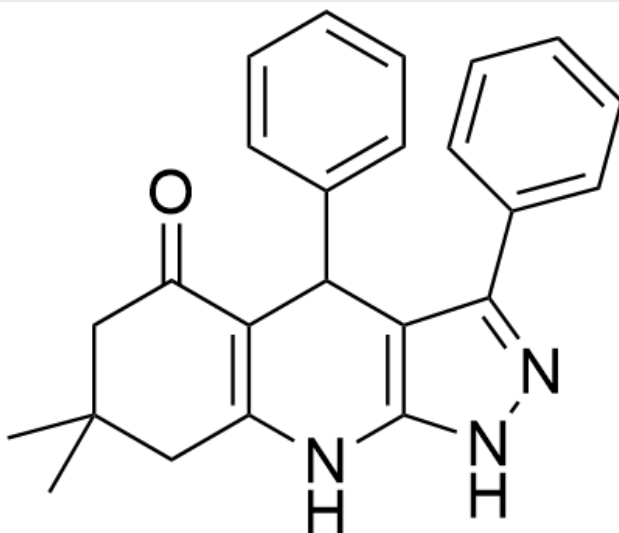
C.



O=C1C2=C(NC3=CC(C4=CC=CC=C4)=NN3C2C5=CC=CC=C5)CC(C)(C)C1

该结构中有8个环

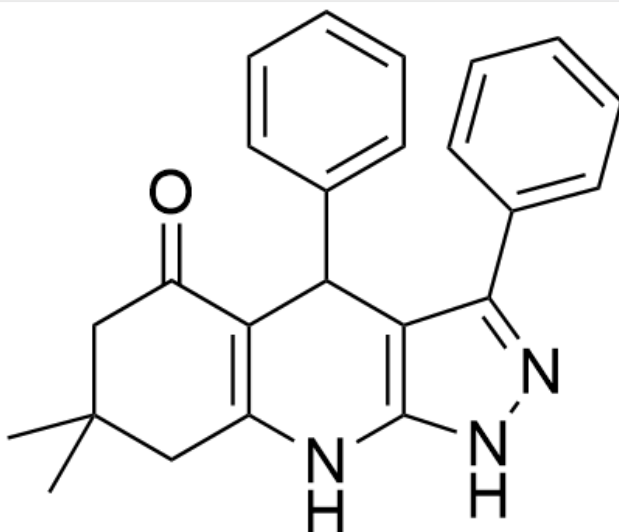
D.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

该结构中有6个环

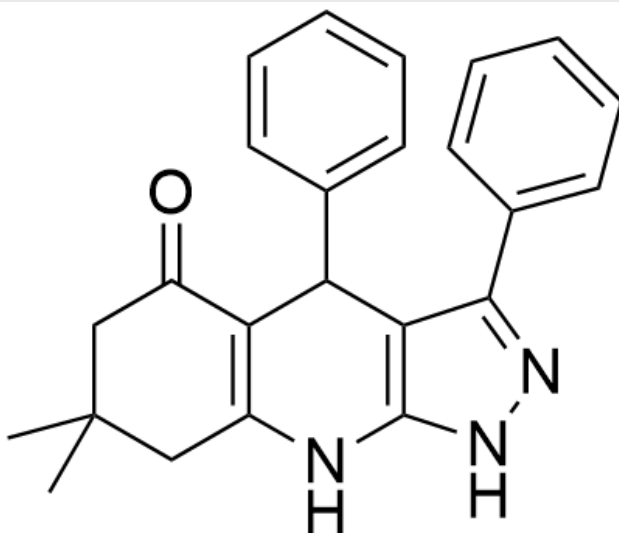
E.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

该结构中有5个环

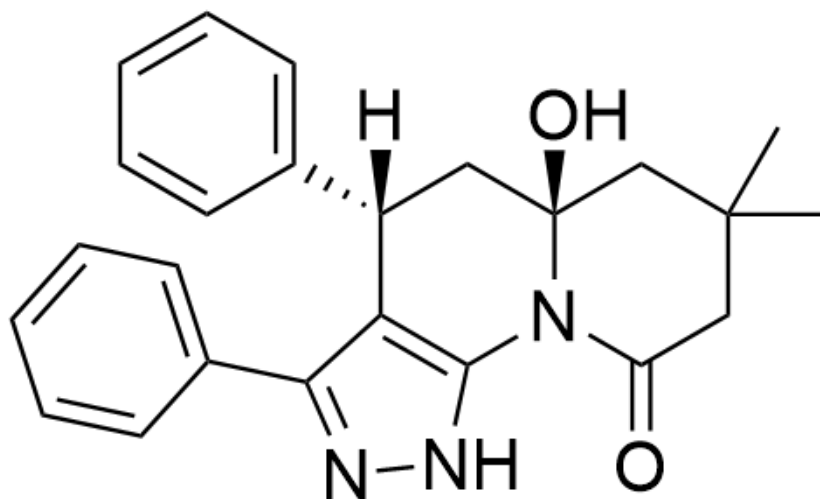
F.



O=C1C2=C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)CC(C)(C)C1

该结构中有8个环

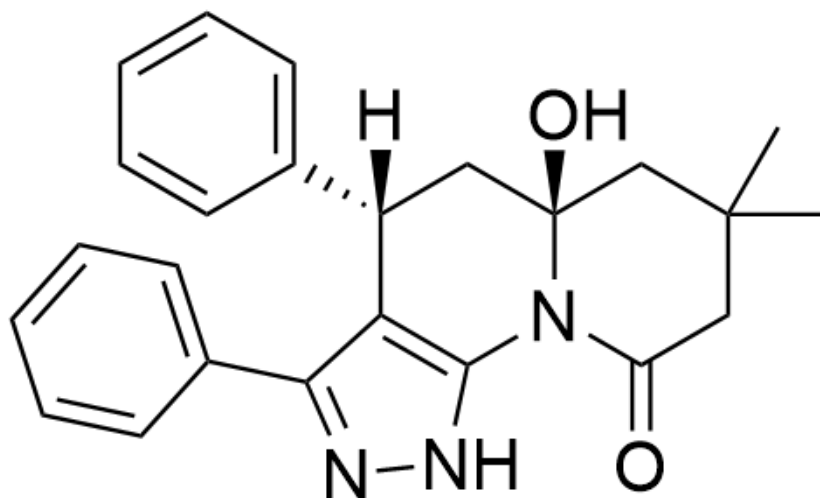
G.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

该结构中有6个环

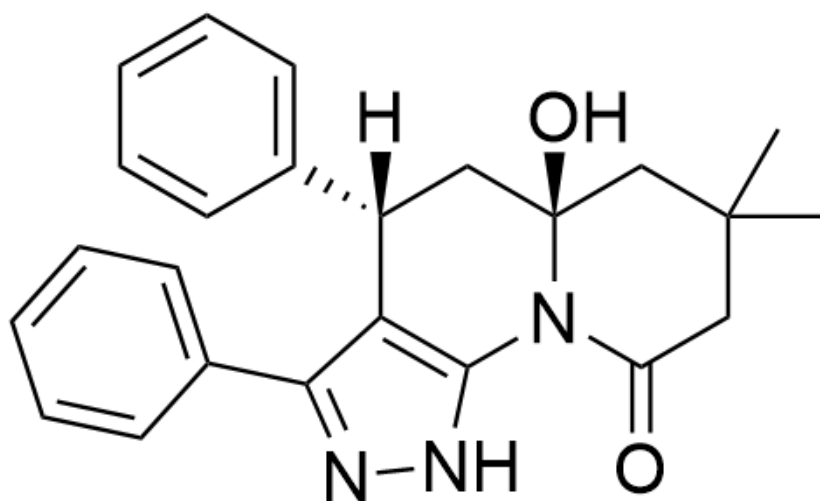
H.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

该结构中有5个环

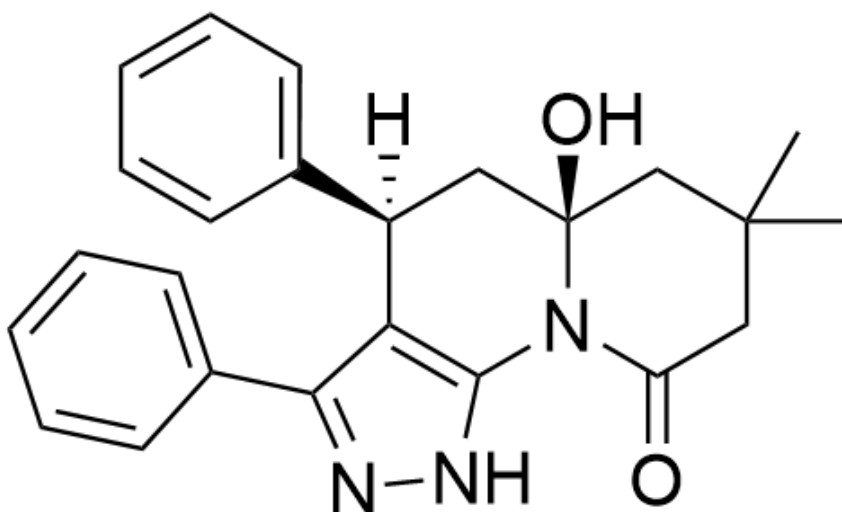
I.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

该结构中有8个环

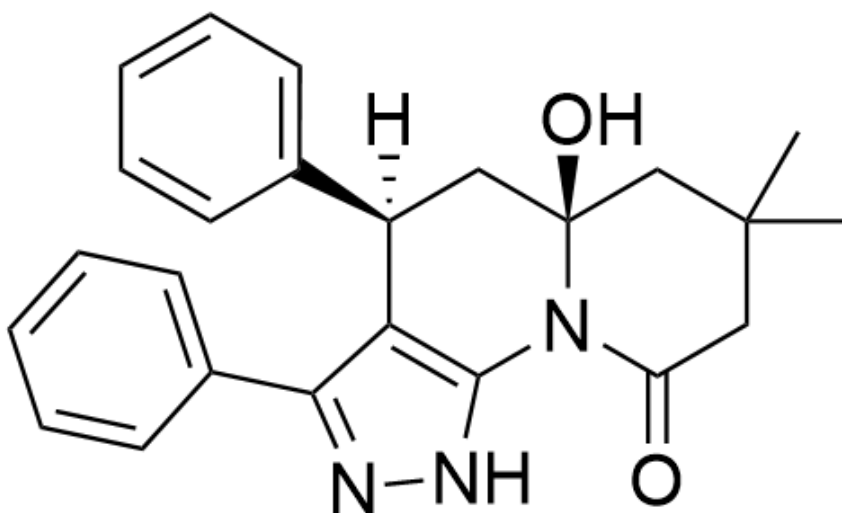
J.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

该结构中有6个环

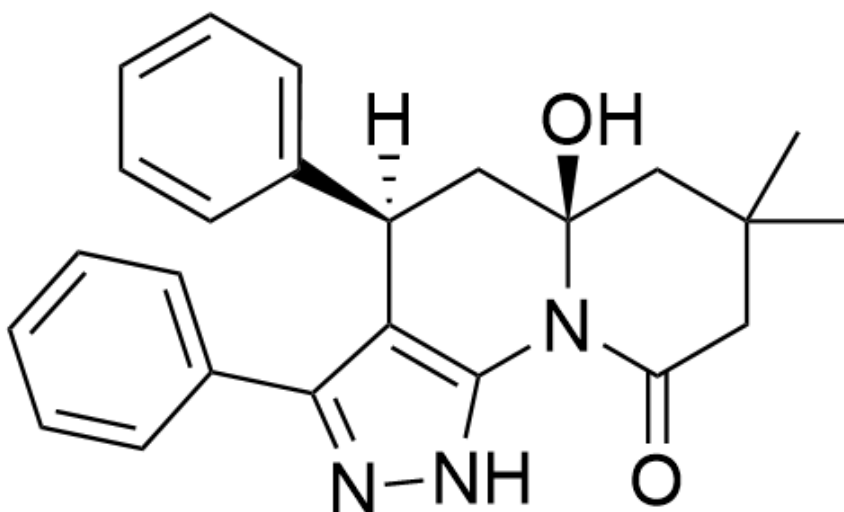
K.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

该结构中有5个环

L.



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1

该结构中有8个环

答案

正确答案: **H**

详细解析

该反应在 150°C 条件下，生成热力学稳定的产物

CHECKPOINT

1 PTS

该反应在 150°C 条件下，生成热力学稳定的产物

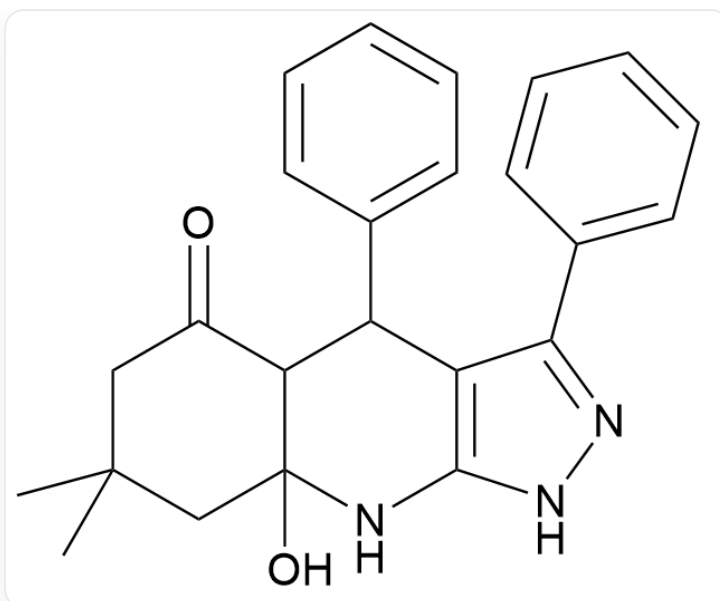
成环时碳碳单键比碳氮单键更加稳定

CHECKPOINT

1 PTS

成环时碳碳单键比碳氮单键更加稳定

因此首先反应形成中间体

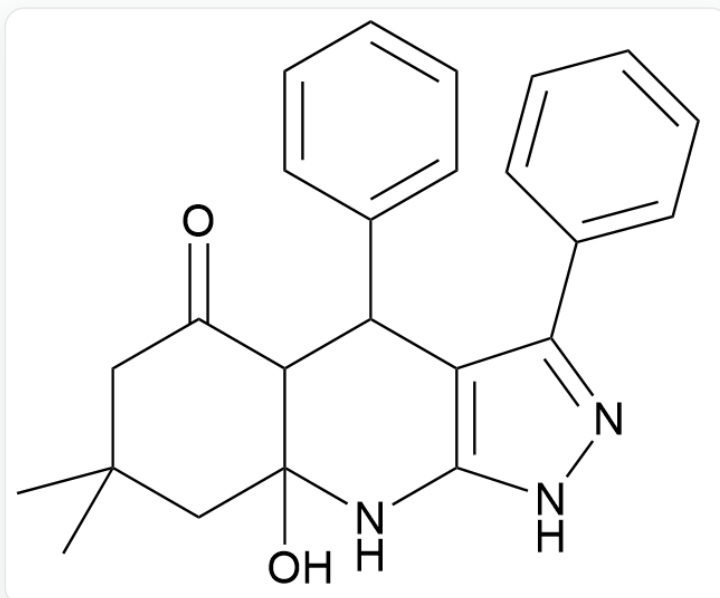


O=C1C2C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)(O)CC(C)(C)C1

CHECKPOINT

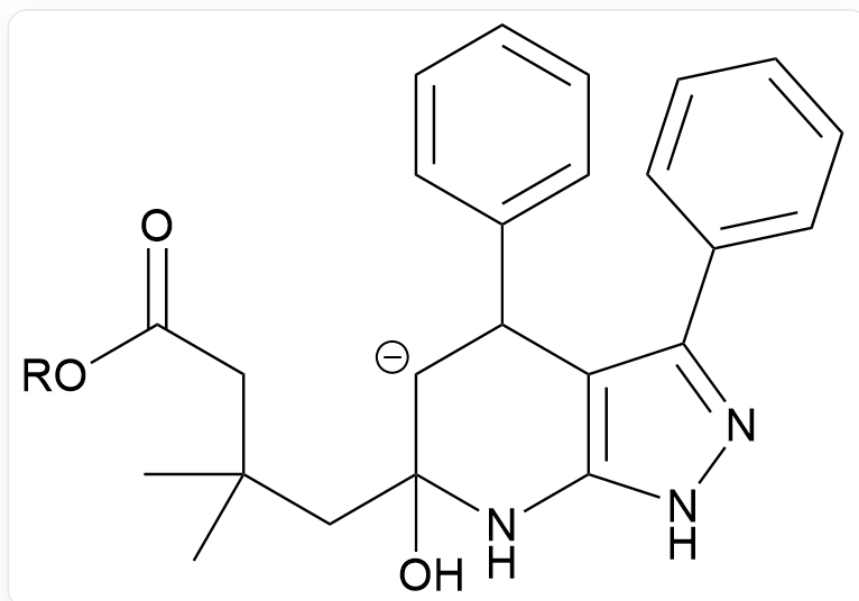
1 PTS

因此首先反应形成中间体



O=C1C2C(NC(NN=C3C4=CC=CC=C4)=C3C2C5=CC=CC=C5)(O)CC(C)(C)C1

随后在强碱的作用下， $tBuO^-$ 或者 EtO^- 会对羰基进行亲核进攻后开环

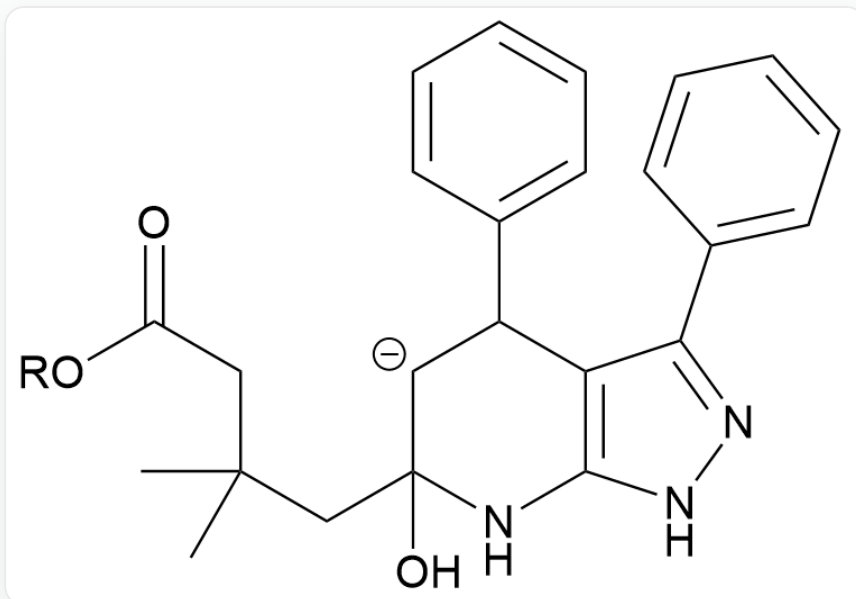


CC(CC1(O)[CH-]C(C2=CC=CC=C2)C3=C(NN=C3C4=CC=CC=C4)N1)(C)CC(O[R])=O

CHECKPOINT

1 PTS

随后在强碱的作用下, $t\text{BuO}^-$ 或者 EtO^- 会对羰基进行亲核进攻后开环



CC(CC1(O)[CH-]C(C2=CC=CC=C2)C3=C(NN=C3C4=CC=CC=C4)N1)(C)CC(O[R])=O

质子交换之后进一步成环形成稳定的内酰胺产物

CHECKPOINT

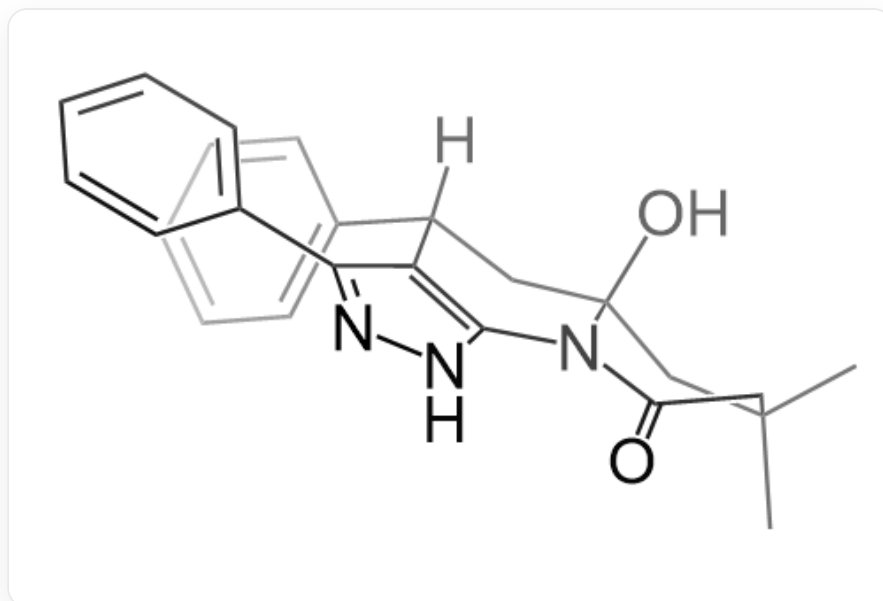
1 PTS

质子交换之后进一步成环形成稳定的内酰胺产物

CHECKPOINT

1 PTS

苯基位于平伏键时为热力学稳定的产物



O=C1N2C3=C(C(C4=CC=CC=C4)=NN3)[C@]([H])(C5=CC=CC=C5)C[C@]2(O)CC(C)(C)C1